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**JASPER: A METHOD, SYSTEM, AND APPARATUS FOR
UNIVERSAL ADAPTIVE INTELLIGENCE ORCHESTRATION**

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This document is submitted as a Provisional Patent Application and does not constitute a non-provisional application. No examination will occur on this filing. The applicant retains the right to file a non-provisional application claiming priority to this filing within twelve (12) months.

SECTION 1 — FIELD OF THE INVENTION

The present invention relates generally to adaptive intelligence orchestration systems; multi-modal cognitive processing apparatus; universal bridge architectures for heterogeneous inference pipeline integration; and seed-initialized computational systems for distributed reasoning workloads.

More particularly, the present invention relates to a unified reduction-to-practice apparatus designated JASPER (Joint Adaptive Semantic Processing and Execution Runtime), which provides a method, system, and apparatus for orchestrating multiple heterogeneous inference subsystems through a universal bridge architecture, a deterministic seed computing initialization framework, and a multi-stage reasoning and inference pipeline, including but not limited to integration with the AETHER X-3 class of operational environments.

SECTION 2 — BACKGROUND OF THE INVENTION

The field of adaptive intelligence orchestration has experienced rapid growth, yet the prior art is marked by significant technical deficiencies that limit the practical deployment of unified, multi-source cognitive processing architectures. The following problems in the prior art are recognized and addressed by the present invention:

Monolithic Architecture Constraints. Existing inference systems in the prior art are monolithic in nature, tightly coupled to a single underlying processing model or computational framework. Such systems are incapable

of dynamically orchestrating multiple heterogeneous inference engines in real-time without substantial re-engineering. This architectural rigidity imposes unacceptable constraints on scalability, adaptability, and operational resilience.

Absence of a Universal Bridge Architecture. No known prior art discloses a universal bridge capable of simultaneously translating, routing, and arbitrating between disparate inference subsystems, heterogeneous data sources, external APIs, sensor arrays, and actuator networks in a bidirectional, real-time manner. Prior bridging mechanisms are point-to-point, protocol-specific, and incapable of generalized cross-subsystem semantic translation.

Lack of Deterministic Reproducibility at the Orchestration Layer. Existing inference pipelines in the prior art lack any mechanism for seed-initialized deterministic reproducibility at the orchestration layer. Stochastic processes within downstream inference subsystems are initialized independently, making forensic audit, session replay, and controlled experimental divergence practically infeasible at the system level.

Absence of a Unified Multi-Modal, Multi-Agent Apparatus. Prior systems do not support apparatus-level integration of multi-modal, multi-agent cognitive pipelines — encompassing text, structured data, code, sensor telemetry, and binary protocol streams — under a single unified reduction-to-practice framework. Existing approaches require bespoke integration layers for each input modality or inference agent, resulting in fragmentation, increased failure surface, and operational inconsistency.

Incompatibility with AETHER X-3 Class Operational Environments. The AETHER X-3 class of operational environments, which requires persistent situational awareness, multi-domain data fusion, sub-50ms adaptive response generation, and tamper-evident audit trails, has no known

compatible prior art apparatus. Existing orchestration systems are not designed for mission-critical, high-availability, multi-classification operational deployments of this character.

There is accordingly a need in the art for a method, system, and apparatus that overcomes the foregoing deficiencies.

SECTION 3 — SUMMARY OF THE INVENTION

The present invention provides a method, system, and apparatus referred to herein as JASPER (Joint Adaptive Semantic Processing and Execution Runtime) for universal adaptive intelligence orchestration. JASPER constitutes a reduction-to-practice of a unified cognitive pipeline apparatus comprising: (1) a Universal Bridge subsystem; (2) a Universal Reasoning and Inference Bridge (URIB) pipeline; (3) a Seed Computing initialization architecture; (4) a multi-LLM orchestration layer; and (5) integration with the AETHER X-3 operational embodiment.

In one embodiment, the apparatus is implemented as a hardware-coupled runtime executing on one or more processors, wherein the hardware-coupled runtime directly interfaces with physical compute nodes, memory subsystems, and network fabric components to provide high-throughput, low-latency orchestration of heterogeneous inference pipelines.

In another embodiment, the apparatus operates within distributed edge-compute nodes, enabling deployment in environments where centralized compute infrastructure is unavailable, impractical, or operationally inadvisable. In this embodiment, the JASPER apparatus is instantiated across a plurality of geographically or logically distributed processing nodes, each

executing a subset of the URIB pipeline stages and coordinating via the Universal Bridge subsystem.

In a further embodiment, the apparatus is instantiated within the AETHER X-3 operational environment, representing a high-capability, mission-critical deployment configuration comprising hardened orchestration infrastructure, FIPS 140-3 compliant Seed Computing, and a curated registry of mission-specialized inference nodes. The AETHER X-3 embodiment is described in detail in Section 4.6 herein.

The present invention, in all of its embodiments, constitutes a reduction-to-practice of a universal adaptive intelligence orchestration apparatus that is novel, non-obvious, and not disclosed by any known prior art as of the filing date of this provisional application.

SECTION 4 — DETAILED DESCRIPTION OF EMBODIMENTS

The following detailed description sets forth specific embodiments of the JASPER apparatus. It is to be understood that the following description is illustrative and not exhaustive. Equivalent structures, methods, and variations within the scope of the present invention will be apparent to persons skilled in the relevant art.

4.1 — JASPER CORE ARCHITECTURE

The JASPER apparatus, in its fundamental embodiment, comprises a plurality of tightly integrated hardware and firmware subsystems that collectively implement a universal adaptive intelligence orchestration capability. The core architecture of the JASPER apparatus includes the following principal components:

(a) Central Orchestration Engine (COE). The COE is the primary computational and control authority of the JASPER apparatus, implemented in hardware-executable instruction sets on one or more processors. The COE coordinates all subsystem activity, manages session lifecycle, enforces operational policies, and interfaces with all other components of the apparatus. The COE is designed for high availability and supports active-active redundancy configurations in multi-node deployments.

(b) Context Memory Bus (CMB). The CMB provides persistent state management across inference sessions, maintaining a shared memory substrate accessible to all active subsystems of the JASPER apparatus. The CMB enables continuity of context across pipeline stages, supports multi-turn inference sessions, and provides the foundational memory architecture upon which session-level determinism is built. The CMB is implemented as a high-bandwidth, low-latency memory fabric with transactional consistency guarantees.

(c) Multi-Modal Input Parser (MMIP). The MMIP is the ingress component of the JASPER apparatus, capable of receiving, parsing, validating, and normalizing inputs across a plurality of modalities. Supported input modalities include, without limitation: natural language text; structured and semi-structured data payloads; executable and interpretable code; sensor telemetry streams; and binary protocol streams conforming to standard and proprietary wire formats. The MMIP normalizes all ingested inputs into JASPER's Internal Semantic Representation Format (ISRF) prior to downstream routing.

(d) Unified Output Synthesis Engine (UOSE). The UOSE is the egress component of the JASPER apparatus, responsible for receiving result envelopes from one or more downstream inference subsystems and synthesizing said envelopes into a coherent, deterministic response artifact. The UOSE applies a weighted consensus algorithm (WCA) to arbitrate among

disparate inference outputs, resolving conflicts, normalizing confidence scores, and producing a single authoritative output artifact for emission to the requesting agent or downstream process.

(e) Arbitration Layer. The arbitration layer is a cross-cutting apparatus component that enforces priority queuing, load balancing, and failover logic across all attached inference subsystems. The arbitration layer monitors subsystem health, load metrics, and reliability scores in real-time, dynamically adjusting dispatch decisions to maintain optimal throughput and resilience. In the event of subsystem failure, the arbitration layer executes autonomous failover protocols to maintain pipeline continuity.

4.2 — UNIVERSAL BRIDGE EMBODIMENT

The Universal Bridge is a hardware-and-firmware apparatus that serves as the primary integration and translation substrate of the JASPER system. The Universal Bridge performs the following functions:

(a) Protocol Translation. The Universal Bridge acts as a protocol translation layer between the JASPER orchestration engine and any number of heterogeneous inference subsystems, APIs, data stores, and sensor arrays. The Universal Bridge enables the JASPER apparatus to communicate with external systems regardless of their native wire protocol, data format, or interface specification.

(b) Bidirectional Semantic Translation Protocol (BSTP). The Universal Bridge implements a bidirectional semantic translation protocol (BSTP) that encodes outbound requests from the JASPER apparatus into a normalized instruction envelope and decodes inbound responses from external subsystems into a standardized result schema. The BSTP is designed to be extensible, supporting the addition of new protocol adapters without modification to the core JASPER apparatus.

(c) Plug-and-Play Subsystem Registration. The Universal Bridge supports plug-and-play subsystem registration, whereby new inference nodes, data sources, or actuators may be attached to the Universal Bridge and made available to the JASPER apparatus without modification to the core orchestration engine or URIB pipeline. Registration is accomplished via a standardized capability declaration protocol that conveys subsystem type, interface specification, capability profile, and operational constraints to the Universal Bridge registry.

(d) Bridge State Register (BSR). The Universal Bridge maintains a persistent bridge state register (BSR) that logs all translation events, including request envelopes, response envelopes, protocol mappings, and timing metadata. The BSR enables comprehensive auditability of all Universal Bridge operations, supports rollback to prior bridge states, and enables deterministic replay of any prior translation sequence given the original seed and BSR state.

(e) Physical Realizations. The Universal Bridge is physically realizable in a plurality of hardware embodiments, including without limitation: (a) a software-defined network appliance firmware module executing on programmable network hardware; (b) an FPGA-instantiated translation core, wherein the BSTP logic and BSR are implemented as configurable hardware logic within a field-programmable gate array; or (c) a dedicated ASIC bridge chip, wherein the Universal Bridge functionality is permanently instantiated in custom silicon for maximum throughput and minimum power consumption.

(f) Supported External System Classes. The Universal Bridge enables the JASPER apparatus to interface with external systems including but not limited to: REST APIs, gRPC endpoints, message queues, relational and vector databases, robotic actuators, Internet-of-Things (IoT) sensor meshes, and satellite telemetry feeds. This breadth of compatibility is a direct

consequence of the BSTP's generalized semantic translation architecture and constitutes a novel contribution of the present invention.

4.3 — URIB PIPELINE (UNIVERSAL REASONING AND INFERENCE BRIDGE)

The Universal Reasoning and Inference Bridge (URIB) pipeline is a structured, multi-stage apparatus comprising six (6) sequential processing stages. The URIB pipeline is the reduction-to-practice core of the JASPER apparatus. It is the operative mechanism by which multi-source, multi-modal inference is unified into a single deterministic output. Each stage of the URIB pipeline is described in detail below:

Stage 1 — Ingestion and Normalization. Raw inputs received at the Multi-Modal Input Parser (MMIP) are tokenized, validated for structural and semantic integrity, and normalized into JASPER's Internal Semantic Representation Format (ISRF). The ISRF is a platform-neutral, schema-validated data structure that provides a uniform representation of any supported input modality, enabling all downstream pipeline stages to operate on a consistent data substrate regardless of the original input type or format.

Stage 2 — Seed Initialization. The ISRF payload generated in Stage 1 is stamped with a deterministic seed vector (DSV) generated by the Seed Computing architecture (described in detail in Section 4.4). The DSV is embedded into the ISRF payload header as a tamper-evident, cryptographically derived initialization token. This seed stamping operation ensures that all downstream stochastic processes within the URIB pipeline and its attached inference subsystems are initialized from a shared, reproducible seed state, enabling deterministic reproducibility of any prior inference session given the original DSV.

Stage 3 — Routing and Dispatch. The arbitration layer evaluates the seeded ISRF payload against a routing policy table (RPT), which encodes the

dispatch logic for all registered inference subsystems. Based on the task ontology classification encoded in the DSV and the capability profiles registered in the dynamic model registry (DMR), the arbitration layer dispatches the seeded payload to one or more downstream inference subsystems via the Universal Bridge. Dispatch may be unicast, multicast, or broadcast depending on the routing policy applicable to the given task class.

Stage 4 — Parallel Inference. One or more inference subsystems receive the dispatched ISRF payload and execute their respective inference processes concurrently. Each subsystem returns a result envelope containing the following standardized fields: (i) a confidence score representing the subsystem's assessed reliability of its output; (ii) a latency metric recording end-to-end processing time; (iii) a token attribution map identifying the input elements most influential to the output; and (iv) a normalized output payload conforming to the result schema of the BSTP.

Stage 5 — Aggregation and Consensus. The Unified Output Synthesis Engine (UOSE) receives all result envelopes returned by dispatched inference subsystems and applies the weighted consensus algorithm (WCA) to synthesize disparate inference results into a single authoritative output artifact. The WCA weights each result envelope by the dispatching subsystem's reliability score, confidence score, and task-model alignment metric, producing a consensus output that maximizes fidelity while minimizing the influence of low-confidence or anomalous results.

Stage 6 — Output Emission and Audit Logging. The synthesized output artifact produced by the UOSE in Stage 5 is emitted to the requesting agent or downstream process via the appropriate output channel. Simultaneously, all pipeline events — including stage transitions, routing decisions, result envelopes, consensus computations, and the final output artifact — are written to an immutable audit log (IAL). The IAL is a cryptographically

chained append-only data structure that provides tamper-evident, forensically admissible records of all JASPER apparatus operations.

4.4 — SEED COMPUTING ARCHITECTURE

The Seed Computing architecture of the JASPER apparatus constitutes a novel initialization paradigm in which a deterministic seed vector (DSV) is generated at the onset of each inference session and propagated throughout the URIB pipeline and all attached inference subsystems. Seed Computing constitutes a novel apparatus-level contribution not disclosed in any known prior art as of the filing date of this provisional application. The Seed Computing architecture is described in detail as follows:

(a) DSV Generation. At the commencement of each inference session, the Seed Computing subsystem generates a deterministic seed vector (DSV). The DSV encodes the following components: (i) a session timestamp at nanosecond resolution, derived from a synchronized hardware clock source; (ii) a user identity hash, computed as a cryptographic digest of the authenticated user or agent identity credential; (iii) a task ontology classification token, derived from the MMIP's semantic analysis of the input payload; (iv) a hardware node fingerprint, encoding the unique hardware identity of the JASPER orchestration node processing the session; and (v) a cryptographically derived entropy component, generated by a hardware random number generator (HRNG) or hardware security module (HSM) to introduce non-reproducible entropy that is recorded in the BSR for forensic traceability.

(b) DSV Propagation. The DSV is injected into the URIB pipeline at Stage 2 and propagated through all downstream subsystems such that each subsystem's stochastic processes are initialized from the shared, reproducible seed state encoded in the DSV. This propagation mechanism ensures end-to-end determinism across the full JASPER apparatus for any given DSV.

(c) Capabilities Enabled by Seed Computing. The Seed Computing architecture enables the following novel operational capabilities: (i) deterministic reproducibility — any prior inference session may be identically reproduced given the original DSV and the BSR state at the time of the session; (ii) controlled divergence experiments — individual DSV components may be modified in isolation to produce controlled variations in inference outcomes, enabling systematic sensitivity analysis; and (iii) forensic auditability — any inference output produced by the JASPER apparatus is traceable to a specific, unique DSV, enabling forensic attribution of outputs to specific sessions, users, hardware nodes, and time instances.

(d) Hardware Embodiment of DSV Generator. The DSV generator may be implemented as a dedicated hardware security module (HSM) integrated with the JASPER orchestration node, or as a trusted platform module (TPM)-integrated seed engine wherein the TPM provides both the hardware identity attestation component and the cryptographic entropy source. In the AETHER X-3 embodiment, the DSV generator is implemented as a FIPS 140-3 compliant HSM subsystem, as described in Section 4.6.

4.5 — MULTI-LLM ORCHESTRATION LAYER

The multi-LLM orchestration layer is an apparatus-level component of JASPER that manages the registration, scheduling, load balancing, and output arbitration of multiple large language model inference nodes (LLM-INS) as subordinate processing units within the JASPER apparatus. In the context of the present invention, each LLM-IN is treated as an interchangeable inference compute resource — a black-box processing unit that accepts a normalized instruction envelope from the Universal Bridge and returns a result envelope conforming to the BSTP result schema. The multi-LLM orchestration layer comprises the following elements:

(a) Dynamic Model Registry (DMR). The orchestration layer maintains a dynamic model registry (DMR) listing all available LLM-INS registered with

the JASPER apparatus. Each DMR entry records: the LLM-IN identifier; capability profile (including supported task classes, input modalities, and output formats); current load metrics; historical reliability scores; and operational status. The DMR is updated in real-time as LLM-INs are registered, deregistered, or their operational status changes.

(b) Capability-Weighted Dispatch Algorithm (CWDA). The arbitration layer selects LLM-INs for each task via a capability-weighted dispatch algorithm (CWDA) that optimizes dispatch decisions across the following dimensions: task-model alignment (the degree to which the LLM-IN's capability profile matches the task ontology classification); latency (the LLM-IN's historical and current response latency metrics); cost (the computational or financial cost of utilizing the LLM-IN); and reliability (the LLM-IN's historical accuracy and uptime metrics). The CWDA produces a ranked dispatch plan that is executed by the arbitration layer.

(c) Multi-Path Inference and Consensus. When the routing policy table specifies multi-path inference for a given task class, the arbitration layer dispatches the seeded payload to multiple LLM-INs concurrently. Each LLM-IN executes its inference process independently and returns a result envelope to the UOSE. The UOSE applies the WCA to produce a consensus output from the set of returned result envelopes. Multi-path inference increases output fidelity, reduces the impact of individual LLM-IN errors, and provides a basis for confidence scoring of the final output artifact.

(d) Hot-Swapping of LLM-INs. The orchestration layer supports hot-swapping of LLM-INs without interrupting active inference sessions. When a LLM-IN is deregistered or becomes unavailable, the arbitration layer automatically redistributes its workload to remaining registered LLM-INs according to the CWDA, maintaining pipeline continuity without session interruption or data loss.

(e) Supported LLM-IN Embodiments. The multi-LLM orchestration layer is compatible with LLM-IN nodes instantiated in a plurality of deployment configurations, including without limitation: cloud-hosted LLM-IN nodes executing on remote server infrastructure; on-premise GPU cluster LLM-INS deployed within the operator's controlled compute environment; edge-deployed quantized LLM-INS executing on resource-constrained edge hardware; and specialized domain-fine-tuned LLM-INS optimized for specific task classes such as legal reasoning, scientific analysis, or code generation.

4.6 — AETHER X-3 EMBODIMENT

The AETHER X-3 embodiment is a specific operational environment instantiation of the JASPER apparatus, designated AETHER X-3, representing a high-capability, mission-critical deployment architecture. The AETHER X-3 embodiment demonstrates reduction-to-practice of the JASPER apparatus in a fully integrated, operationally validated configuration. The AETHER X-3 embodiment comprises the following principal components and distinguishing features:

(a) Principal Components. The AETHER X-3 embodiment comprises: (i) a dedicated JASPER orchestration node cluster configured for active-active high availability; (ii) an AETHER-class Universal Bridge array with redundant failover, providing continuous translation and routing services across all registered inference subsystems; (iii) a hardened URIB pipeline with a sub-50ms end-to-end latency target from input ingestion to output emission; (iv) a secure Seed Computing HSM subsystem compliant with FIPS 140-3 standards for cryptographic module security; and (v) a curated registry of mission-specialized LLM-INS selected and validated for the operational requirements of the AETHER X-3 environment.

(b) Operational Design Requirements. The AETHER X-3 embodiment is specifically designed for operational environments requiring: persistent situational awareness across multiple concurrent inference sessions; multi-

domain data fusion integrating inputs from heterogeneous sensor, intelligence, and command sources; real-time adaptive response generation within the sub-50ms latency target; and tamper-evident audit trails providing forensically admissible records of all apparatus operations.

(c) Key Distinguishing Features of AETHER X-3. The AETHER X-3 embodiment is distinguished from general JASPER deployments by the following advanced capabilities:

- **Autonomous Pipeline Reconfiguration.** The AETHER X-3 embodiment implements autonomous pipeline reconfiguration logic within the COE, enabling the apparatus to detect and respond to subsystem failures by dynamically restructuring the URIB pipeline, redistributing workloads, and maintaining operational continuity without operator intervention.
- **Cryptographically Signed Output Artifacts.** All output artifacts emitted by the AETHER X-3 embodiment are cryptographically signed using a hardware-bound signing key maintained in the FIPS 140-3 compliant HSM subsystem, providing non-repudiable provenance attestation for every output produced by the apparatus.
- **Multi-Classification Data Handling.** The AETHER X-3 Universal Bridge array implements compartmentalized bridge segments, each operating within a defined classification boundary, enabling the apparatus to simultaneously process data of multiple classification levels without cross-contamination between compartments.
- **C2 Interface Integration.** The AETHER X-3 embodiment provides integration with external command-and-control (C2) interfaces via standardized military messaging protocols, enabling the JASPER apparatus to receive tasking directives from and emit output artifacts to external C2 systems within the operational environment.

SECTION 5 — CLAIMS (PROVISIONAL — REPRESENTATIVE)

NOTE:

The following claims are representative and non-exhaustive. These claims are presented for the purpose of establishing the scope of the inventive concept and satisfying the disclosure requirements of a provisional patent application under 35 U.S.C. § 111(b). Claims will be formalized, refined, and expanded in the corresponding non-provisional application filed pursuant to 35 U.S.C. § 111(a).

1. A method for universal adaptive intelligence orchestration comprising: (a) receiving a multi-modal input at a multi-modal input parser (MMIP) of a JASPER apparatus; (b) normalizing said input into an internal semantic representation format (ISRF); (c) generating a deterministic seed vector (DSV) via a seed computing subsystem of the JASPER apparatus; (d) stamping the ISRF payload with the DSV to produce a seeded payload; (e) routing the seeded payload to one or more inference subsystems via a universal bridge of the JASPER apparatus; (f) receiving result envelopes from said one or more inference subsystems; and (g) synthesizing said result envelopes into a unified output artifact via a unified output synthesis engine (UOSE) of the JASPER apparatus.
2. A system for universal adaptive intelligence orchestration comprising: a central orchestration engine (COE); a context memory bus (CMB) communicatively coupled to the COE; a universal bridge apparatus communicatively coupled to the COE; a universal reasoning and inference bridge (URIB) pipeline operatively implemented by the COE; a seed computing architecture operatively coupled to the URIB pipeline; a multi-

LLM orchestration layer managed by the COE; and an AETHER X-3 operational embodiment instantiating all of the foregoing components in a mission-critical deployment configuration.

3. An apparatus for universal adaptive intelligence orchestration comprising one or more processors executing instruction sets that implement: a universal bridge configured to translate between heterogeneous inference subsystems using a bidirectional semantic translation protocol (BSTP); a universal reasoning and inference bridge (URIB) pipeline configured to route, process, and synthesize inference outputs through a plurality of sequential processing stages; and a seed computing architecture configured to generate and propagate deterministic seed vectors (DSVs) for reproducible initialization of inference sessions.
4. The method of claim 1, wherein the universal bridge performs protocol translation between the JASPER apparatus and at least one heterogeneous inference subsystem by encoding outbound requests into a normalized instruction envelope and decoding inbound responses into a standardized result schema pursuant to the bidirectional semantic translation protocol (BSTP).
5. The method of claim 4, wherein the BSTP normalized instruction envelope comprises: a task ontology classification token; a seeded ISRF payload; a routing policy reference; a session identifier; and one or more capability requirements specifying constraints on the receiving inference subsystem.
6. The apparatus of claim 3, wherein the universal bridge maintains a persistent bridge state register (BSR) that records all translation events, including outbound instruction envelopes, inbound result envelopes, protocol mapping tables applied, and nanosecond-resolution timing

metadata, thereby enabling auditability, rollback, and deterministic replay of any prior translation sequence.

7. The apparatus of claim 3, wherein the URIB pipeline comprises six sequential processing stages consisting of: Stage 1 — Ingestion and Normalization; Stage 2 — Seed Initialization; Stage 3 — Routing and Dispatch; Stage 4 — Parallel Inference; Stage 5 — Aggregation and Consensus; and Stage 6 — Output Emission and Audit Logging.
8. The apparatus of claim 7, wherein Stage 1 of the URIB pipeline tokenizes, validates, and normalizes multi-modal inputs received at the MMIP into the internal semantic representation format (ISRF), and wherein supported input modalities include at least: natural language text, structured data, executable code, sensor telemetry streams, and binary protocol streams.
9. The apparatus of claim 7, wherein Stage 3 of the URIB pipeline evaluates the seeded ISRF payload against a routing policy table (RPT) and dispatches the payload to one or more registered inference subsystems, and wherein dispatch mode is selected from the group consisting of: unicast, multicast, and broadcast, based on the routing policy applicable to the task ontology class of the payload.
10. The method of claim 1, wherein the deterministic seed vector (DSV) encodes: (a) a session timestamp at nanosecond resolution derived from a synchronized hardware clock source; (b) a user identity hash computed as a cryptographic digest of an authenticated user or agent identity credential; (c) a task ontology classification token derived from semantic analysis of the input payload; (d) a hardware node fingerprint encoding the unique hardware identity of the processing node; and (e) a cryptographically derived entropy component generated by a hardware entropy source.

- 11.** The apparatus of claim 3, wherein the cryptographically derived entropy component of the DSV is generated by a hardware security module (HSM) integrated with the one or more processors, and wherein the HSM is compliant with FIPS 140-3 standards for cryptographic module security.
- 12.** The apparatus of claim 3, wherein the cryptographically derived entropy component of the DSV is generated by a trusted platform module (TPM)-integrated seed engine, wherein the TPM provides hardware identity attestation for the hardware node fingerprint component and a hardware random number source for the entropy component.
- 13.** The apparatus of claim 7, wherein Stage 5 of the URIB pipeline applies a weighted consensus algorithm (WCA) to synthesize result envelopes received from a plurality of inference subsystems into a single authoritative output artifact, and wherein the WCA weights each result envelope by the dispatching subsystem's reliability score, confidence score, and task-model alignment metric.
- 14.** The system of claim 2, wherein the multi-LLM orchestration layer maintains a dynamic model registry (DMR) recording, for each registered large language model inference node (LLM-IN): a unique identifier; a capability profile including supported task classes and input modalities; current load metrics; historical reliability scores; and operational status.
- 15.** The system of claim 14, wherein the multi-LLM orchestration layer selects LLM-INS for dispatch via a capability-weighted dispatch algorithm (CWDA) that optimizes dispatch decisions across the dimensions of task-model alignment, latency, cost, and reliability, and produces a ranked dispatch plan executed by the arbitration layer.
- 16.** The system of claim 2, wherein the multi-LLM orchestration layer supports hot-swapping of registered LLM-INS without interrupting active inference sessions, and wherein the arbitration layer automatically

redistributes workloads from a deregistered LLM-IN to remaining registered LLM-INS according to the CWDA upon detection of LLM-IN unavailability.

- 17.** The system of claim 2, wherein the AETHER X-3 operational embodiment comprises a dedicated JASPER orchestration node cluster configured for active-active high availability, an AETHER-class Universal Bridge array with redundant failover, a hardened URIB pipeline with a sub-50ms end-to-end latency target, a FIPS 140-3 compliant Seed Computing HSM subsystem, and a curated registry of mission-specialized LLM-INS.
- 18.** The system of claim 17, wherein the AETHER X-3 operational embodiment implements autonomous pipeline reconfiguration logic within the COE, enabling the apparatus to detect subsystem failures and dynamically restructure the URIB pipeline to maintain operational continuity without operator intervention.
- 19.** The system of claim 17, wherein all output artifacts emitted by the AETHER X-3 operational embodiment are cryptographically signed using a hardware-bound signing key maintained in the FIPS 140-3 compliant HSM subsystem, providing non-repudiable provenance attestation for every output artifact.
- 20.** The system of claim 17, wherein the AETHER X-3 Universal Bridge array implements compartmentalized bridge segments each operating within a defined data classification boundary, enabling simultaneous processing of data of multiple classification levels without cross-contamination between compartments.
- 21.** The apparatus of claim 3, wherein the universal bridge is physically realized as a field-programmable gate array (FPGA)-instantiated translation core, wherein the BSTP logic and bridge state register (BSR) are implemented as configurable hardware logic within the FPGA.

- 22.** The apparatus of claim 3, wherein the universal bridge is physically realized as a dedicated application-specific integrated circuit (ASIC) bridge chip, wherein the universal bridge functionality is permanently instantiated in custom silicon.
- 23.** The apparatus of claim 7, wherein Stage 6 of the URIB pipeline writes all pipeline events to an immutable audit log (IAL) implemented as a cryptographically chained, append-only data structure, thereby providing tamper-evident, forensically admissible records of all JASPER apparatus operations.
- 24.** The system of claim 17, wherein the AETHER X-3 operational embodiment provides integration with one or more external command-and-control (C2) interfaces via standardized military messaging protocols, enabling receipt of tasking directives from and emission of output artifacts to external C2 systems.
- 25.** The apparatus of claim 3, wherein the one or more processors are distributed across a plurality of geographically or logically separated edge-compute nodes, each executing a subset of the URIB pipeline stages and coordinating via the universal bridge.
- 26.** The method of claim 1, wherein reproducibility of any prior inference session is achieved by re-executing the method with the original DSV from the prior session, such that all stochastic processes within the URIB pipeline and attached inference subsystems are re-initialized from the same seed state, producing a deterministically identical output artifact.
- 27.** The apparatus of claim 3, wherein the universal bridge supports plug-and-play subsystem registration, whereby new inference nodes, data sources, or actuators are registered with the universal bridge via a standardized capability declaration protocol without modification to the core orchestration engine or URIB pipeline.

SECTION 6 — ABSTRACT

A method, system, and apparatus designated JASPER (Joint Adaptive Semantic Processing and Execution Runtime) provides universal adaptive intelligence orchestration through a unified reduction-to-practice architecture. The JASPER apparatus comprises a Universal Bridge subsystem implementing a bidirectional semantic translation protocol for interfacing with heterogeneous inference subsystems, data stores, and sensor arrays; a Universal Reasoning and Inference Bridge (URIB) pipeline executing a six-stage process of ingestion, seed initialization, routing, parallel inference, consensus aggregation, and audit logging; a Seed Computing architecture generating deterministic seed vectors for reproducible, forensically auditable inference sessions; a multi-LLM orchestration layer managing registration, dispatch, and output arbitration of multiple inference nodes via a capability-weighted dispatch algorithm; and an AETHER X-3 operational embodiment representing a mission-critical, FIPS 140-3 compliant, high-availability instantiation of the full apparatus. The invention constitutes a reduction-to-practice of a unified adaptive intelligence orchestration architecture.

SECTION 7 — BRIEF DESCRIPTION OF DRAWINGS

NOTE:

Actual drawings to be submitted with the non-provisional application. The following descriptions identify the figures to be provided.

The accompanying drawings, which are to be submitted with the non-provisional application, illustrate embodiments of the invention and, together with the Detailed Description of Embodiments in Section 4, serve to explain the principles of the JASPER apparatus. In the drawings:

FIG. 1 — High-level block diagram of the JASPER apparatus, depicting the central orchestration engine (COE), context memory bus (CMB), multi-modal input parser (MMIP), unified output synthesis engine (UOSE), arbitration layer, Universal Bridge, URIB pipeline, Seed Computing architecture, and multi-LLM orchestration layer, and their interconnections.

FIG. 2 — Universal Bridge protocol translation architecture, depicting the bidirectional semantic translation protocol (BSTP) encoding and decoding pathways, the bridge state register (BSR), the plug-and-play subsystem registration interface, and representative external system connections including REST APIs, gRPC endpoints, message queues, vector databases, IoT sensor meshes, and satellite telemetry feeds.

FIG. 3 — URIB Pipeline six-stage data flow diagram, depicting the sequential flow of data through Stage 1 (Ingestion and Normalization), Stage 2 (Seed Initialization), Stage 3 (Routing and Dispatch), Stage 4 (Parallel Inference), Stage 5 (Aggregation and Consensus), and Stage 6 (Output Emission and Audit Logging), including the data structures present at each stage transition.

FIG. 4 — Seed Computing deterministic seed vector (DSV) generation architecture, depicting the DSV generator hardware embodiment, the five DSV component inputs (session timestamp, user identity hash, task ontology classification, hardware node fingerprint, and cryptographic entropy component), the DSV assembly and injection mechanism at Stage 2 of the URIB pipeline, and the HSM/TPM integration pathway.

FIG. 5 — AETHER X-3 full system embodiment block diagram, depicting the dedicated orchestration node cluster, AETHER-class Universal Bridge array with redundant failover, hardened URIB pipeline, FIPS 140-3 compliant HSM subsystem, curated LLM-IN registry, autonomous pipeline reconfiguration logic, cryptographic output signing mechanism, compartmentalized bridge segments, and C2 interface integration pathways.

INVENTOR DECLARATION

I hereby declare that I am the original inventor of the subject matter claimed in this provisional patent application and that the information contained herein is true and correct to the best of my knowledge, information, and belief.

Signature of Inventor Leon [Redacted Signature]	Date July 18, 2026
Residence: Briarcliff, Texas, United States of America Filing Time: 20:40 CDT (Central Daylight Time)	

PROVISIONAL PATENT APPLICATION DISCLAIMER

This provisional patent application establishes a priority date under 35 U.S.C. § 111(b). The applicant has 12 months from this filing date to file a corresponding non-provisional application to claim the benefit of this provisional filing date. This provisional application will not be examined, will not issue as a patent, and will be abandoned automatically if a corresponding non-provisional application is not timely filed. The filing of this provisional application does not initiate the 20-year patent term.

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